OPEN ARCHITECTURE STANDARD · v0.1.0

EASRA

Enterprise AI Systems Reference Architecture

The open standard for production-grade Enterprise AI.

VENDOR-NEUTRAL
No lock-in. Every layer maps to ≥ 2 clouds.

PRODUCTION-READY
Runtime, deployment & scaling — not a slideware framework.

VERIFICATION-FIRST
Grounding, citation, factuality baked in — verification ≠ evaluation.

CLOUD-AGNOSTIC
Azure · AWS · GCP · OSS-K8s — same architecture, four impls.

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Enterprise AI stacks are one-offs.

Every team is redesigning the same architecture — badly, in isolation, with different vocabulary.

No shared vocabulary
Teams argue about ‘agent’ vs ‘workflow’ vs ‘orchestrator’ before writing a line of code.

Security bolted on
Prompt injection, tool safety and PII treated as add-ons, not architectural concerns.

Verification confused with evaluation
Offline metrics ship to prod. No first-class verification of grounding, citation, factuality.

Cost & observability afterthoughts
Token spend, latency SLOs and tool traces retrofitted after the outage.

An open architecture standard for Enterprise AI.

Vendor-neutral. Production-grade. Standards-aligned.

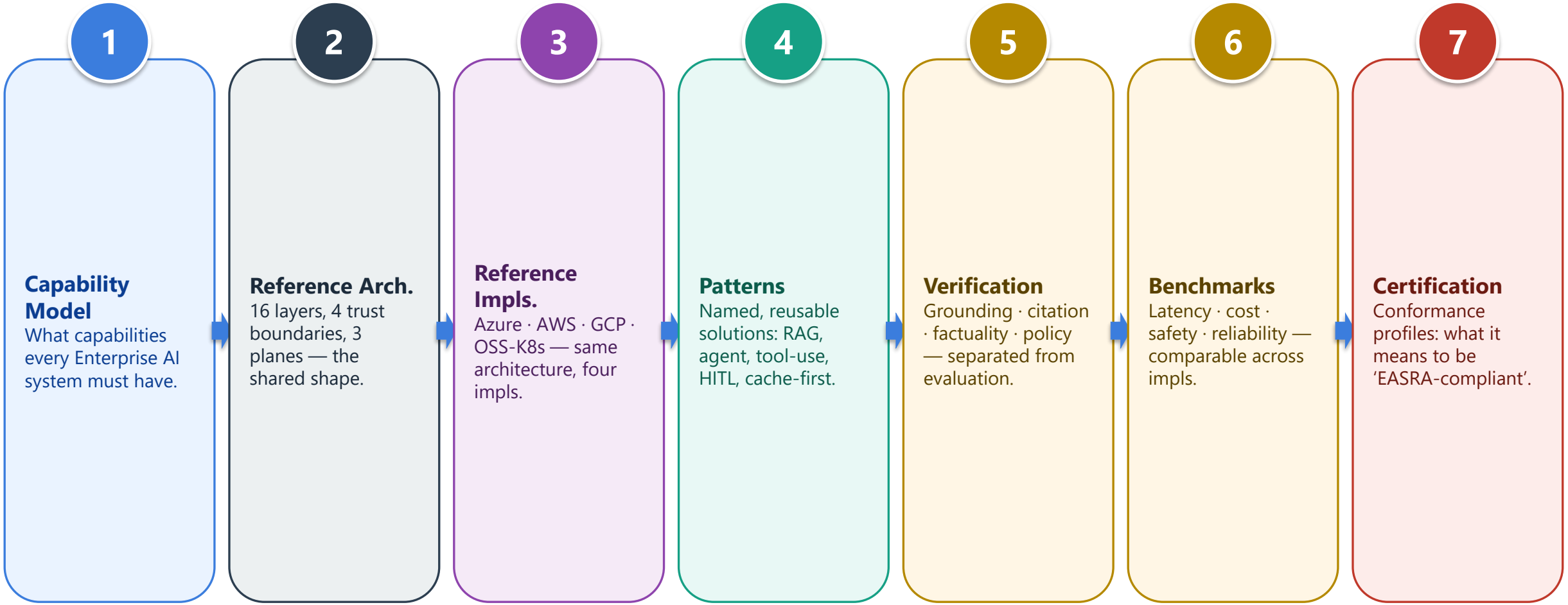
Vendor-neutral
Every layer maps cleanly to Azure, AWS, GCP and open-source. No lock-in.

Security by design
Zero trust, four trust boundaries, prompt-injection resistance in the layers, not on top.

Verification-first
Grounding, citation, factuality, policy and safety verification separated from evaluation.

Cost + observability by default
Tokens, latency, cost, prompts, tools traced through OpenTelemetry from every component.

EASRA is not just a diagram. It is a full stack — like TOGAF, like OpenTelemetry.



Ladders 1–3 are v0.1.0 today. Ladders 4–7 are the roadmap.

P1 **Vendor Neutrality**
Logical layers; each maps to ≥ 2 independent implementations.

P2 **Security by Design**
Zero-trust, least-privilege, prompt-injection resistance are architectural.

P3 **Verification by Design**
Every AI output is verifiable; verification ≠ evaluation.

P4 **Observability by Default**
Traces, metrics, logs, token/cost accounting emitted from every component.

P5 **Loose Coupling**
Layers communicate only through published interfaces.

P6 **Externalized State**
Memory, sessions, caches live outside compute; compute is stateless.

P7 **Human-in-the-Loop**
Every autonomous action has escalation, override and audit paths.

P8 **Failure Isolation**
Layer failure degrades gracefully; no single layer cascades a full outage.

P9 **Cost Awareness**
Token, compute, storage cost are explicit design inputs.

P10 **Evolvability**
New models, tools, agents, channels added without redesigning the architecture.

EASRA · The Sixteen Layers

L0 — Channels & User Experience
Web, mobile, API, chat, voice, SDK — every way a user or system talks to the platform.

L1 — Edge · Gateway · Identity
CDN, WAF, global load balancer, API gateway, OIDC/OAuth2, rate limiting, session, request validation.

L2 — AI Orchestration
Router, single agent, multi-agent coordinator, planner/workflow engine.

L3 — Prompt Intelligence
Context builder, prompt builder, versioned prompt registry.

L4 — Memory & Context
Short-term, long-term, session, semantic memory and user profile.

L5 — Knowledge & Retrieval
Retrieval router across vector, BM25, SQL, graph; reranker; hybrid retrieval.

L6 — AI Models & Model Router
Capability/cost/latency/policy-aware routing over foundation + small/specialised models.

L7 — Tooling & Actions (MCP)
Tool router, MCP clients/servers, enterprise APIs, impact-class enforcer, tool registry.

L8 — Guardrails & Safety
Input, prompt, tool-argument and output guardrails.

L9 — Verification
First-class verification layer: grounding, citation, factuality, format, policy, safety.

L10 — Observability
OpenTelemetry-based traces, metrics, logs; tokens, latency, cost; agent/prompt/tool trace; evaluation, alerts.

L11 — Performance · Caching · Cost
Prompt, semantic, embedding, memory, response caches; cost ledger.

L12 — LLMOps & Delivery
Git → CI (lint, sec-scan, prompt-tests, eval) → artefact registry → CD (canary, blue/green, shadow, rollback).

L13 — Security & Zero Trust
AuthN/AuthZ, secrets, encryption, PII detection, policy engine, audit, compliance.

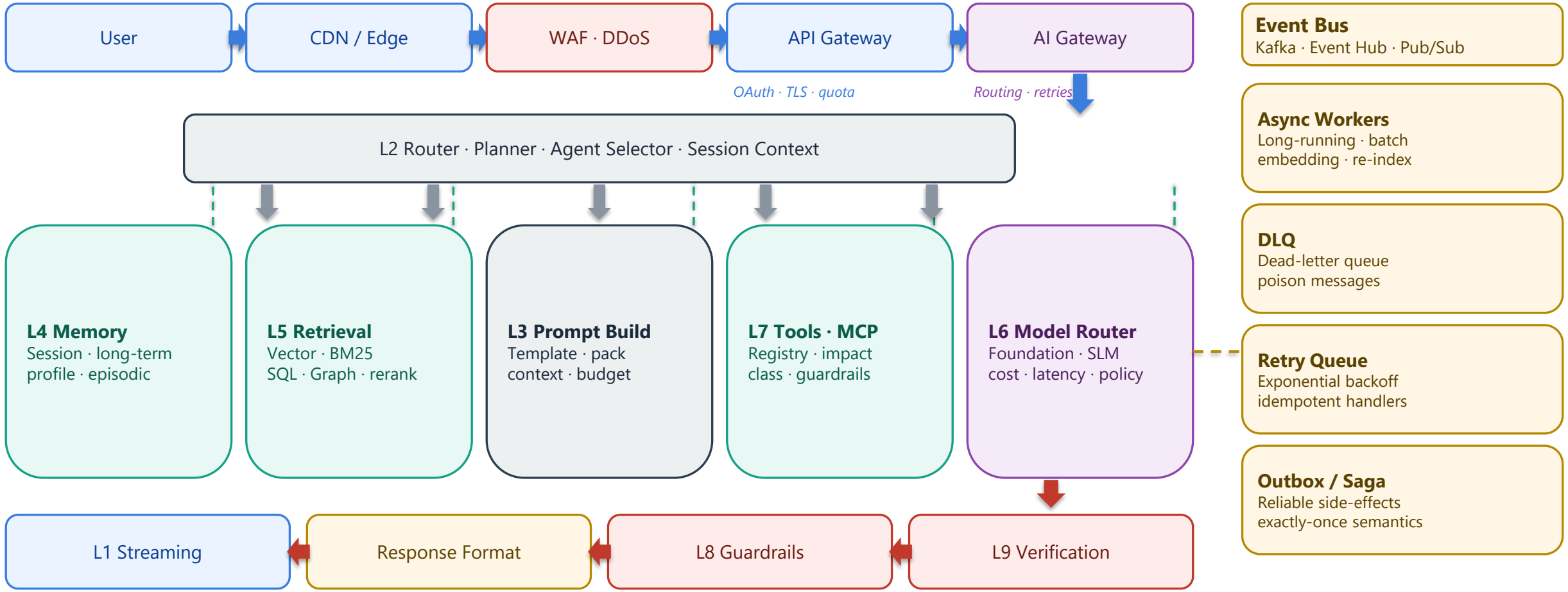
L14 — Governance · Risk · Compliance
Policy engine, model/prompt/tool registries, NIST AI RMF, ISO 42001, EU AI Act, OWASP LLM Top 10.

L15 — Business Outcomes & Value
KPIs, value attribution, cost/benefit, adoption tracking.

EASRA · Enterprise AI Systems Reference Architecture

Production runtime architecture

Request flows down through the pipeline; response streams back through verification and guardrails. Async work fans out to an event bus.



← Streaming response back to User · same pipeline in reverse · citations · redaction · verdict

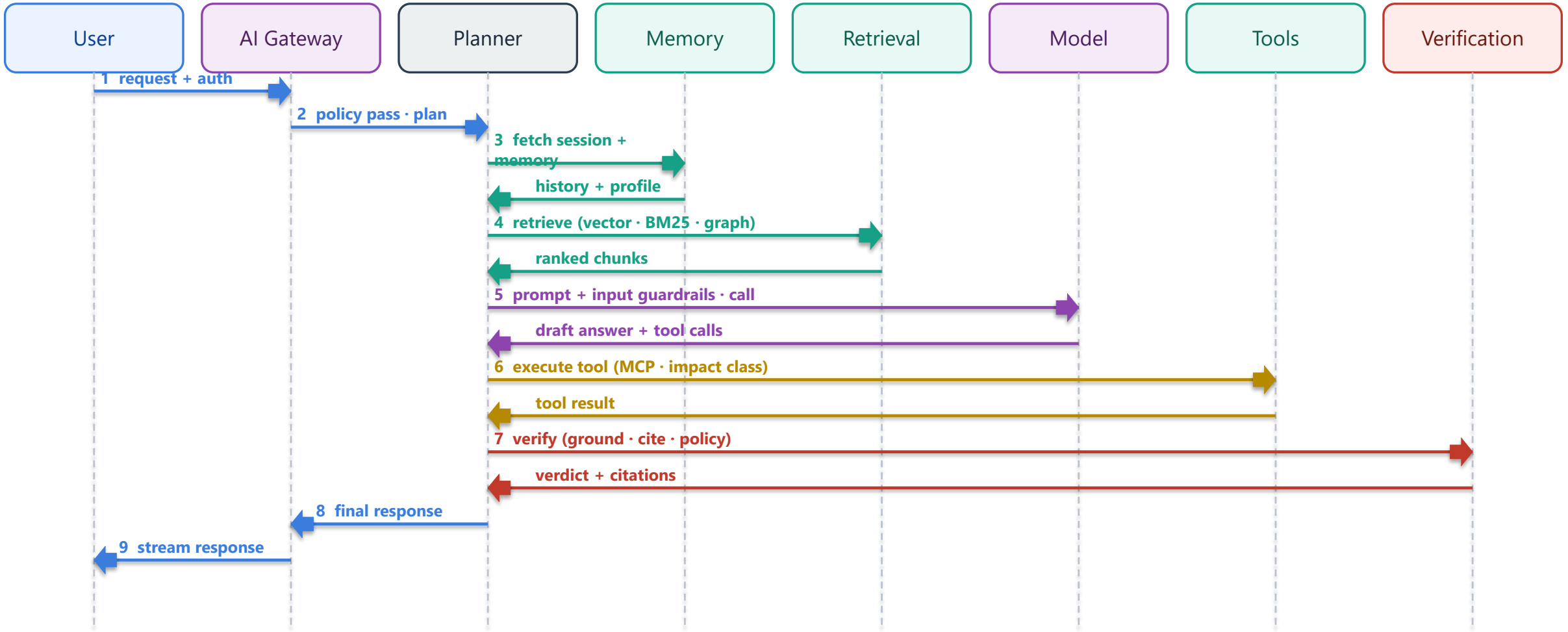
L10 Observability Plane
OpenTelemetry · traces · metrics · logs · token/cost accounting

L13 Security Plane
Zero trust · IAM · secrets · encryption · threat detection · policy

L15 Cost & Value Plane
Token budgets · autoscaling · GPU scheduling · load shedding · backpressure

Solid arrows = request path · Dashed teal = response · Dashed amber = async / eventing · Bottom strip = cross-cutting planes

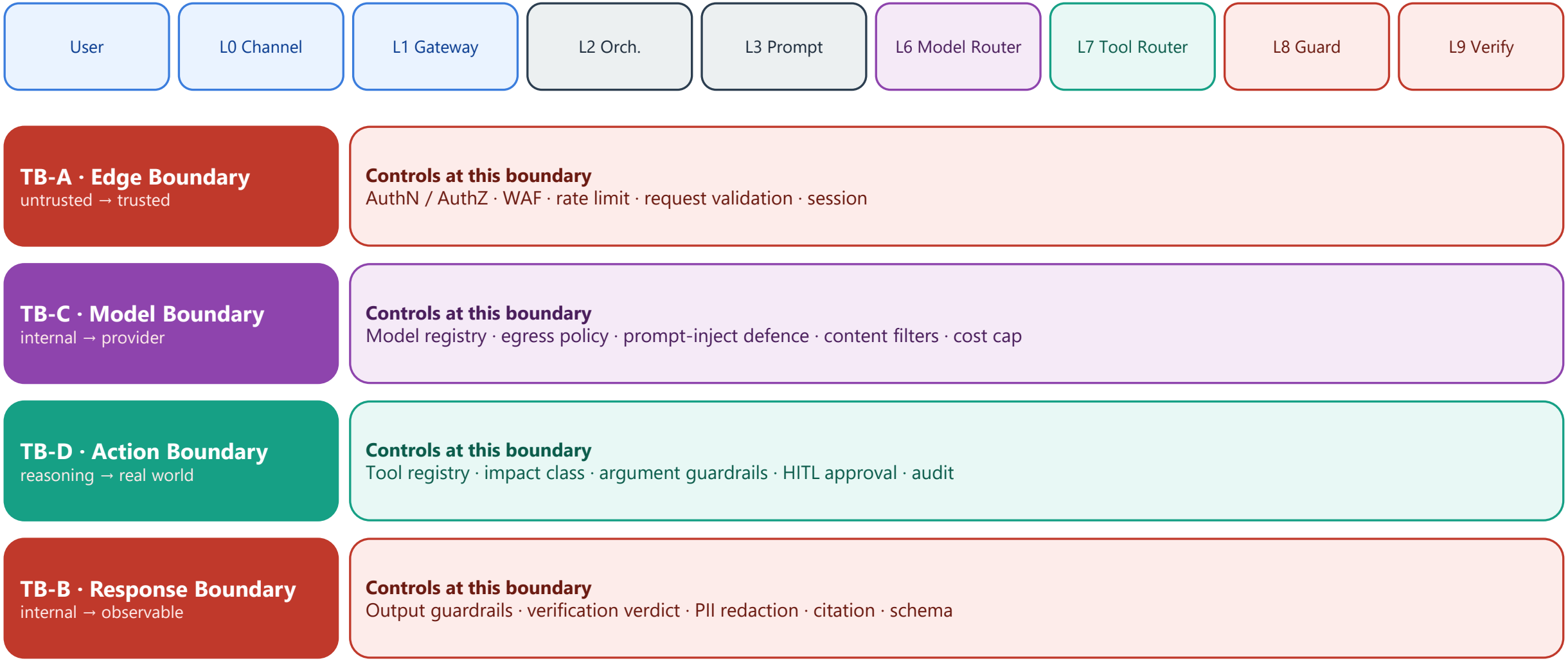
One request across the actors that actually cooperate at runtime. Read top-to-bottom; each arrow is a real hop with real latency.



Sequence diagram · not a layered stack. Every arrow is a hop, a timeout, a retry, an SLO. Solid arrow = call · return arrow = response.

EASRA · Enterprise AI Systems Reference Architecture

Every request crosses four boundaries. Each has its own attackers, controls, and failure modes.



Verification ≠ evaluation. Every model output passes through this pipeline before it reaches the user.

1. **Grounding**
   Is every claim supported by retrieved context?
2. **Citation**
   Are sources attached, valid, resolvable?
3. **Semantic**
   Does the answer address the actual question?
4. **Constraint**
   Structural: schema · format · length · language.
5. **Execution**
   Generated code / SQL executes without harm.
6. **Policy**
   Enterprise policy · role · jurisdiction · data class.
7. **Safety**
   Toxicity · self-harm · CSAM · disallowed content.
8. **Business Rules**
   Pricing · discount · escalation · SLA guardrails.
9. **Confidence**
   Calibrated uncertainty score · abstain threshold.
10. **Approval**
    HITL for high-impact · async review queue.

Any stage may VETO · abstain · escalate to HITL · downgrade confidence · request re-generation.

Enterprise AI has three planes — just like Kubernetes. Confusing them is how governance, cost and reliability go wrong.

Control Plane

Decides · governs · configures

- Policy Engine (OPA · Cedar)
- Configuration store
- Model Registry
- Prompt Registry
- Workflow Registry
- Tool Registry
- Feature Flags
- Model routing rules
- Rate-limit + quota rules

Data Plane

Runs · executes · serves traffic

- Prompts & context assembly
- Embeddings & retrieval
- Model inference calls
- Tool executions (MCP)
- Verification checkers
- Guardrail evaluators
- Streaming responses
- Cache reads / writes
- Event bus messages

Management Plane

Observes · pays · improves

- CI / CD pipelines
- OpenTelemetry traces & metrics
- Cost dashboards & budgets
- Alerting & on-call
- Governance & audit logs
- Compliance evidence
- LLMOps eval & benchmarks
- Incident response
- Capacity + quota planning

Between the AI reasoning layer and cloud infrastructure sits a platform layer of shared, stateful services. Enterprise AI stands or falls on this middle.

L11 Cache Manager
Prompt · semantic · embedding
memory · response caches

Kafka / Event Bus
Async pipelines · streaming events

Scheduler
Cron · workflows · DAGs · Airflow / Argo

Configuration
App Config · Consul · git-versioned

Secrets
Key Vault · Vault · KMS · rotation

Feature Flags
Progressive delivery · kill-switches

Workflow Registry
Named agents · SOPs · replayable runs

Prompt Registry
Versioned prompts · A/B · rollback

Policy Engine
OPA · Cedar · content + tool policies

Artifact Registry
Signed containers · SBOM · attestations

Model Registry
Foundation · fine-tuned · shadow · prod

Tool Registry
MCP servers · impact class · rate limit

EASRA · Enterprise AI Systems Reference Architecture

Logical architecture ≠ deployment architecture. Here is what actually runs, autoscales and pages someone at 3 AM.

Internet → Front Door / CDN → WAF · DDoS · Bot → Private Endpoint

AKS · Container Apps · (or ECS · GKE · K8s) — HPA + KEDA + GPU scheduler + PodDisruptionBudget

Gateway pods
APIM sidecar
5–50 replicas
HPA on RPS

Orchestrator pods
Container Apps
3–30 replicas
HPA on queue

Agent workers
GPU nodepool
2–20 replicas
KEDA on Kafka

Tool executors
Sandboxed
5–30 replicas
rate limited

Verifier pods
CPU nodepool
3–15 replicas
HPA on backlog

Redis Enterprise
cache + rate-limit
zone-redundant

Qdrant / AI Search
vector index
replicated

Cosmos DB
session + memory
multi-region

Azure OpenAI
AOAI · Foundry
provisioned + PAYG

Blob Storage
docs · attachments
lifecycle rules

OpenTelemetry
Collector →
App Insights / Grafana

Same shape on AWS (EKS + ElastiCache + OpenSearch + Bedrock + S3 + CloudWatch) and GCP (GKE + Memorystore + Vertex + GCS + Cloud Ops).

Traditional CI/CD is not enough. LLMOps adds prompt tests, eval gates, model registry, canary + shadow, cost guardrails.

Git / Repo
Prompts · agents · tools · IaC · eval sets
versioned as code

CI
Lint · sec-scan · unit · prompt tests
safety tests · eval gate

Artefact Registry
Signed containers · pinned models
prompt versions · SBOM

CD
Blue / Green · Canary · Shadow
Rollback · progressive delivery

Runtime
Continuous eval · drift · cost cap
incident + rollback triggers

Prompt lifecycle

Draft → Test → Approved → Prod → Deprecated

- Versioned prompt templates
- Prompt tests + regression suite
- Prompt registry as source of truth
- Prompt rollback independent of model

Model lifecycle

Candidate → Shadow → Canary → Prod → Retired

- Model registry (foundation + fine-tuned)
- Approved providers only (TB-C)
- Cost + latency + safety benchmarks
- Shadow traffic for regression

Tool lifecycle

Onboarded → Restricted → GA → Deprecated

- Tool registry with impact class
- Argument guardrails per tool
- HITL approval for high-impact
- Audit + rate limit per tool

Same 16 layers. Four independent mappings. Vendor-neutral by construction.

Microsoft Azure

L1 Gateway
APIM · Front Door · Entra ID
L2 Orch.
Container Apps · Durable Fn
L5 Retrieval
AI Search · Cosmos Gremlin · Fabric
L6 Models
Azure OpenAI · AI Foundry
L7 Tools
APIM · Logic Apps · MCP on ACA
L10 Obs.
App Insights · Azure Monitor
L13 Sec.
Entra · Key Vault · Purview · Defender
L14 GRC
Purview · Azure Policy · Foundry gov.

Amazon AWS

L1 Gateway
CloudFront · WAF · API Gateway · Cognito
L2 Orch.
ECS / Fargate · Step Functions
L5 Retrieval
OpenSearch · Neptune · Athena
L6 Models
Bedrock · SageMaker
L7 Tools
API Gateway · Lambda · MCP on ECS
L10 Obs.
CloudWatch · X-Ray · OTel Collector
L13 Sec.
IAM · KMS · Secrets Manager · Macie
L14 GRC
Config · Audit Manager · Bedrock Guardrails

Google Cloud

L1 Gateway
Cloud LB · Cloud Armor · API Gateway · IAP
L2 Orch.
Cloud Run · Workflows
L5 Retrieval
Vertex AI Search · Spanner Graph
L6 Models
Vertex AI · Gemini · Model Garden
L7 Tools
API Gateway · Cloud Functions · MCP on Run
L10 Obs.
Cloud Ops Suite · OTel Collector
L13 Sec.
IAM · KMS · Secret Manager · SCC
L14 GRC
Assured Workloads · Vertex Model Registry

Open Source / K8s

L1 Gateway
Envoy · Kong · Keycloak · OIDC
L2 Orch.
K8s + Argo Workflows · LangGraph
L5 Retrieval
Qdrant · Weaviate · pgvector · Neo4j
L6 Models
vLLM · Ollama · TGI · TorchServe
L7 Tools
MCP servers · custom adapters
L10 Obs.
OTel · Prometheus · Grafana · Tempo · Loki
L13 Sec.
SPIFFE · Vault · Falco · Kyverno
L14 GRC
OPA / Gatekeeper · Kyverno · Sigstore

EASRA maps explicitly to the frameworks your compliance, security and risk teams already recognise.

NIST AI RMF
Govern · Map · Measure · Manage
Mapped in EASRA at: L14 Governance · L10 Observability

MITRE ATLAS
Adversary tactics + techniques for AI
Mapped in EASRA at: L1 · L8 Guardrails · L13 · L14

ISO / IEC 42001
AI management system
Mapped in EASRA at: L14 · L12 LLMOps · L15 Value

OpenTelemetry
Traces / metrics / logs schema
Mapped in EASRA at: L10 Observability across every layer

EU AI Act
Risk categorisation + provider duties
Mapped in EASRA at: L14 · L9 Verification · L13 Security

Model Context Protocol (MCP)
Standardised tool interface
Mapped in EASRA at: L7 Tooling & Actions

OWASP LLM Top 10
LLM-specific application risks
Mapped in EASRA at: L1 · L6 · L7 · L8 · L9 · L11

TOGAF
Enterprise architecture context
Mapped in EASRA at: Positions EASRA as an AI-domain reference architecture

Every consequential architectural choice in EASRA is recorded as an ADR. Not a deliverable — a first-class artefact of the standard.

ADR-001
Verification is separate from Evaluation
Runtime verifiers vs. offline eval — different lifecycles, different owners.

ADR-002
Memory is externalised
Compute stays stateless. Sessions, profiles, episodic memory live outside.

ADR-003
Prompt Registry is mandatory
Prompts are versioned code, not literals — enables A/B, rollback, audit.

ADR-004
AI Gateway is separate from API Gateway
Model routing, token budgets, retries, circuit breakers ≠ HTTP concerns.

ADR-005
Tool Registry with impact class is mandatory
Every tool has an impact class; HITL for high-impact; audit for all.

ADR-006
Cache Manager is centralised
One component for prompt · semantic · embedding · memory · response caches.

ADR-007
Async work uses an Event Bus, not sync calls
Batch, embedding, re-index, long-running agents — Kafka / Event Hub.

ADR-008
Control / Data / Management planes are separated
Governance, runtime and operations scale independently.

ADR-009
Standards are mapped, not replaced
NIST · ISO 42001 · EU AI Act · OWASP LLM · MITRE ATLAS — complements, doesn't compete.

ADR-010
Dual license: CC-BY-4.0 docs · Apache-2.0 code
Docs freely adaptable; code freely usable in commercial products.

ADRs live under docs/adr/. Each one has context · decision · consequences · status · date.

ADOPT · CONTRIBUTE · FORK

EASRA is an open architecture standard.

Vendor-neutral. Standards-aligned. Verifiable. Yours to fork.

★ Star
github.com/KarthikeyanDhanakotti/EASRA

Fork & Adopt
templates/repository-template/

Contribute
CONTRIBUTING.md · GOVERNANCE.md

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